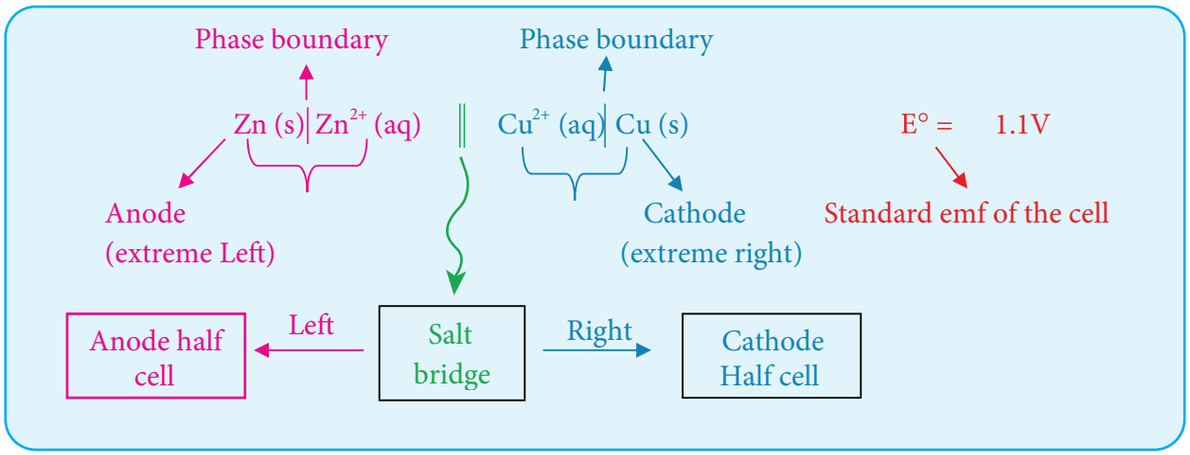


Example

The net redox reaction of a galvanic cell is given below

$2\,Cr\,(s) + 3Cu^{2+}(aq) \rightarrow 2Cr^{3+}(aq) + 3Cu\,(s)$

Write the half reactions and describe the cell using cell notation.

Anodic oxidation $2Cr\,(s) \rightarrow 2Cr^{3+}\,(aq) + 6e^{-}$ .....(1)

Cathodic reduction : $3Cu^{2+}\,(aq) + 6e^{-} \rightarrow 3\,Cu\,(s)$ .....(2)

Cell Notation is

$Cr\,(s)\,|\,Cr^{3+}(aq)\,\|\,Cu^{2+}(aq)\,|\,Cu(s)$

9.3.3 *emf* of a Cell

We have learnt that when two half cells of a Daniel cell are connected, a spontaneous redox reaction will take place which results in the flow of electrons from anode to cathode. The force that pushes the electrons away from the anode and pulls them towards cathode is called the electromotive force (emf) (or) the cell potential. The SI unit of cell potential is the volt (v).

When there is one volt difference in electrical potential between the anode and cathode, one joule of energy is released for each columb of charge that moves between them.

i.e., $1J = 1C \times 1V$ .....(9.18)

The cell voltage depends on the nature of the electrodes, the concentration of the electrolytes and the temperature at which the cell is operated. For example

At, $25^{\circ}C$, The emf of the below mentioned Daniel cell is 1.107 Volts

$Zn\,(s)\,|\,Zn^{2+}(aq,1M)\,\|\,Cu^{2+}(aq,1M)\,|\,Cu\,(s)$ $E^{0} = 1.107\ V$

XII U9 Electro Chemistry.indd 47 11-01-2021 14:44:07